

**To what extent do ESG (environmental, social, and governance)  
disclosures affect investor behavior in global financial markets?**

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## Introduction

“88% of global investors say they are interested in sustainable investing, with more than half very interested.” — Morgan Stanley Sustainable Signals Report.

In today’s financial landscape, the traditional pursuit of profit is no longer sufficient. Investors are increasingly factoring in a company’s broader impact on the world - environmentally, socially and in terms of governance. The accelerating urgency of climate change and a demand for corporate transparency have propelled environmental, social and governance considerations to the center of investment and corporate strategy. Companies are no longer judged solely by their earnings, but by how responsibly they operate within the natural and social ecosystems that surround them.

This shift is reinforced by a rising tide of regulatory involvement. ESG disclosures, once voluntary, are now becoming mandatory in many jurisdictions. Stock exchanges and financial regulators across the globe, particularly in the United States and the European Union, are pushing companies to report on their ESG performance especially around climate-related risks, emissions and governance practices. As a result, more than 90% of S&P 500 companies and a growing number of firms worldwide now publish annual ESG reports.

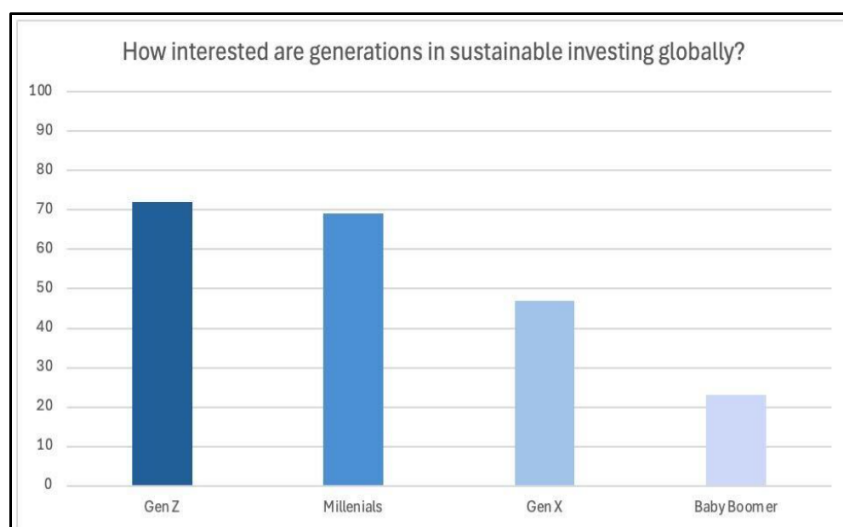
At its core, ESG is a strategic framework that enables stakeholders to assess how an organization manages sustainability-related risks and opportunities. ESG goes beyond traditional CSR or compliance-driven models. It integrates environmental, social and governance dimensions into business operations in a measurable, systematic, and accountable way.

The Environmental pillar evaluates a company’s impact on the natural world using tracking indicators like greenhouse gas emissions, energy usage, waste management and climate

risk preparedness. Social disclosures focus on how a company treats people, whether through employee conditions, human rights, community impact or supply chain ethics. This is especially relevant in countries with weaker labor protections. The Governance component examines internal corporate structures by assessing board diversity, executive compensation, shareholder rights and the ethical integrity of decision-making. Together, these three pillars provide a 360-degree view of how a company interacts with the world around it.

Typical ESG disclosures include carbon footprint data, DEI efforts, employee training hours, ethical supply chain practices and governance controls like anti-corruption policies. While traditional financial reporting focuses only on economic performance, ESG disclosures aim to capture the sustainability and societal impact of corporate action.

Sustainable finance has grown as a result of financial investors' attention being drawn to responsible investments (RI) in recent decades due to a notable shift in social and environmental consciousness. Sustainable investing incorporates ESG factors—environmental, social, and governance—into investment schemes. Sustainable investors take into account how businesses or projects affect society, the environment, and their own governance procedures in addition to financial performance. Aligning financial objectives with the overarching goal of bringing about a beneficial social or environmental change is the aim.



(Figure Source: [Investing in the future: The rise of sustainable finance | UN Global Compact](#))

According to the Morgan Stanley report, 80 percent of young investors plan to increase their sustainable investments this year. 72 percent of Gen Z and 69 percent of Millennials are “very interested” in ESG investing, compared to much lower rates among older generations. A growing desire for ethical and sustainable investment is being driven by the beliefs and interests of Gen Z and millennial investors, who are starting to dominate the market. Growing public awareness, activism, and international projects like the Paris Agreement, the Sustainable Development Goals of the UN, and the "Great Wealth Transfer" are all contributing factors to this trend. These younger generations' investing decisions will have a big impact on the financial environment since they stand to inherit trillions of dollars. A growing number of professional programs are responding to this change by providing training in sustainable finance to financial practitioners in order to address the increasing demand.

Long-term thinking is promoted by sustainable investing, which is consistent with more general goals like risk mitigation and building resilience to systemic issues like inequality and climate change. Sustainable finance is a type of risk reduction that improves portfolio durability in a world that is becoming more and more volatile. This approach provides a means of

balancing disturbances. In order to satisfy the needs of a world that is changing quickly, it also offers financial advisors and investment platforms the chance to provide sustainable solutions and counsel. Businesses that ignore ESG considerations run a greater risk of damaging their brand. Transparency is essential in the current digital era because any mistakes can go viral very fast and seriously harm a brand's reputation. Offenses are more difficult to fix, and companies that ignore social or environmental issues frequently face boycotts and public outrage, which can lower shareholder value. Businesses that include ESG standards into their operations not only shield themselves from these dangers but also show that they are taking proactive measures to resolve these issues. As ESG gains traction, banking and finance professionals feel compelled to have a deeper understanding of the ideas in order to make wise decisions for their own companies and enterprises.

In addition, governments everywhere are implementing more stringent laws to encourage sustainability and guarantee preparedness for the future. For example, in order to increase corporate accountability, the U.S. Securities and Exchange Commission (SEC) has increased its examination of ESG disclosures and put more pressure on organizations to implement sustainable business practices. This pattern emphasizes how crucial sustainable financing will be in determining the direction of ethical corporate practices, green infrastructure, and technology in the future. Customers and clients are beginning to put pressure on even small and medium-sized businesses (SMEs) to change their business strategy to incorporate sustainability goals, launch their sustainability journey, and lower their carbon footprint. Investigating resource-efficient manufacturing, sustainability tactics, and ethical issues in design processes are all crucial when it comes to engineering designs and manufacturing procedures.

When investors are making decisions, ESG ratings act as a standard. Companies are rated according to their ESG reporting by rating organizations such as Morningstar and MSCI ESG Research. A company's relative performance within its industry is then indicated by the ESG scores that the agencies assign. These ratings are used by investors to evaluate businesses and find those who have robust ESG policies that support their sustainability goals. Investment choices, portfolio building, and corporate involvement are all influenced by ESG ratings. But it's crucial to understand that ESG ratings have limits because they rely on sustainability reporting, which might differ in terms of consistency and quality. To further understand a company's ESG procedures, investors should perform additional due diligence and view ESG ratings as one tool among many.

### **Conceptual Framework**

*Environmental, social, and governance, or ESG*, is a framework for evaluating a company's ethical effect and sustainability. Investors, customers, and stakeholders must evaluate businesses using an ESG approach. It assesses a company's performance in areas that impact the environment and society in addition to typical financial measurements. This method assists investors in making well-informed choices that complement their long-term objectives and values. The environmental pillar evaluates a company's impact on the natural world using tracking indicators like greenhouse gas emissions, energy usage, waste management, and climate risk preparedness. Social disclosures focus on how a company treats people, whether through employee conditions, human rights, community impact, or supply chain ethics. This is especially relevant in countries with weaker labor protections. The governance component examines internal corporate structures by assessing board diversity, executive compensation, shareholder

rights, and the ethical integrity of decision-making. Together, these three pillars provide a 360-degree view of how a company interacts with the world around it.

Typical ESG disclosures include carbon footprint data, DEI efforts, employee training hours, ethical supply chain practices, and governance controls like anti-corruption policies. While traditional financial reporting focuses only on economic performance, ESG disclosures aim to capture the sustainability and societal impact of corporate action. Several businesses include more detailed metrics in separate sustainability reports, which frequently include an index proving compliance with one or more of the major disclosure frameworks. Many companies also disclose the most important aspects of their ESG governance and strategy in their SEC (U.S. Securities and Exchange Commission) reports, concentrating on qualitative discussions of items that are important to management and strategic planning of the business in accordance with the SEC's guidelines regarding climate change information and non-financial disclosures. Transparency standards have been enhanced by frameworks like the Task Force on Climate-related Financial Disclosures (TCFD) and the EU's Corporate Sustainability Reporting Directive (CSRD).

The term "financial market" refers to a market that acts as a conduit between savers and borrowers by moving cash or funds from individuals with excess funds to those who require funding or investment. To put it simply, a financial market is one that generates and trades financial assets. Generally speaking, business enterprises are referred to as deficit units and investors as surplus units. Therefore, a financial market connects lenders and borrowers and serves as a conduit between surplus and deficit units. In order to comprehend how emotions and cognitive biases affect financial decision-making, behavioral finance integrates psychological ideas with traditional economic and financial theories. By departing from the presumptions of

conventional finance models, this interdisciplinary approach aids in explaining why people frequently make irrational financial decisions. Because it offers important insights into the psychological aspects driving financial decisions, an understanding of behavioral finance is essential for investors, financial professionals, and policymakers. Thus, in this paper we are trying to quantify or show the effect and the extent of the effect of ESG disclosures on investor behavior in global financial markets.

### **ESG and its Evolution**

As we have already seen the definition of ESG. We will now delve deeper into what ESG is. It generally takes three factors into consideration as follows.

***Environmental Factors:*** ESG's environmental components center on how a business engages with the environment. This includes analyzing a business's environmental performance, which entails determining how it affects environmental elements including recycling, the use of eco-friendly materials, water conservation, and carbon emission reduction. Understanding a company's possible financial risks also requires evaluating environmental hazards, such as resource scarcity and greenhouse gas emissions. Businesses that are dedicated to environmental sustainability make the world a healthier place and are better equipped to comply with regulations.

***Social Factors:*** Under environmental, social, and governance, the social component looks at how a business handles its connections with its customers, suppliers, workers, and the community in which it operates. Ethical work practices, consumer happiness, employee welfare, and diversity and inclusion are important issues. A company's reputation can be improved, and stakeholder trust can be increased with a great social performance.

**Governance Factors:** A company's management and leadership structures are part of its governance. It covers shareholder rights, diversity on the board, ethical business practices, and transparency. Accountability is guaranteed by good governance, which lowers the possibility of corruption and improves stability over the long run.



The evolution of ESG can be traced back through several movements.

**EHS (environment, health, and safety):** EHS frameworks have focused on regulating pollution and improving workplace safety since the 1980s. It focused on mitigating harm caused by business operations.

**Corporate sustainability:** EHS is into a broader corporate sustainability movement. This often involved companies setting self-imposed environmental goals. However, critics agreed that it was sometimes used more for marketing than genuine impact. This practice is called “greenwashing”.

**CSR (corporate social responsibility):** In the early 2000s, corporate sustainability combined social issues under the CSR umbrella. This included community engagement, philanthropy, and employee volunteerism. However, CSR lacked robust metrics and usually relied heavily on PR-driven initiatives.

**ESG:** The term “ESG” was popularized in a 2004 UN report titled “Who Cares Wins,” but it wasn’t until the late 2010s and early 2020s that ESG emerged as a systematic and proactive framework. ESG is distinct in that it integrates environmental, social, and governance factors in a measurable and accountable way. This aims to align corporate behavior with stakeholder expectations and long-term sustainability.

Up until recently, its fortunes had been slowly improving since the acronym "ESG" (environmental, social, and governance) was created in 2005. For instance, since 2019, internet searches for ESG have increased fivefold, despite a fall in searches for "CSR" (corporate social responsibility), an earlier focus area that was more indicative of corporate engagement than changes to a core business model. Organizations have been investing more in enhancing ESG across all sectors, regions, and business sizes. Approximately 70% of Russell 1000 companies and over 90% of S&P 500 companies now publish ESG reports in some capacity. Reporting ESG factors is either required or actively being considered in a number of jurisdictions. The Securities and Exchange Commission (SEC) in the US is debating proposed regulations that would mandate more thorough disclosure of greenhouse gas (GHG) emissions and climate-related risks.

2 There are also pending or suggested further SEC regulations on various aspects of ESG.

Even if the number of new investments has been declining recently, the growing importance of ESG in investments has also been clearly visible. For instance, inflows into sustainable funds increased from \$5 billion in 2018 to over \$50 billion in 2020 and then to

almost \$70 billion in 2021. In the first quarter of 2022, these funds received \$87 billion in net new money, and in the second quarter, they received \$33 billion. 4. Global sustainable assets stand at over \$2.5 trillion at the halfway point of 2022. This is less than the market's 14.6 percent decline over the same time period, but it still represents a 13.3 percent decline from the end of Q1 2022. Green bonds, ESG-linked mutual funds, and sustainability exchange-traded funds allow investors to align portfolios with their values, often without sacrificing returns. While some critics argue there is no proven ESG alpha, many investors believe that companies aligned with ESG principles are more resilient and better positioned for long-term value creation. The environmental aspect of ESG and reactions to climate change have been key factors in its rise. However, other aspects of ESG, particularly the social dimension, have also been becoming more and more well-known. According to one analysis, the number of shareholder proposals pertaining to social issues increased by 37% during the 2021 proxy season as compared to the prior year.

Global rating agencies such as MSCI, Sustainalytics, Refinitiv, and S&P Global play a crucial role in shaping perceptions of ESG performance. However, the key challenge is the lack of standardization. ESG scores from different agencies often correlate only around 54%, compared to 99% for credit ratings. Despite this inconsistency, progress is underway. Reporting frameworks are consolidating, especially with the creation of the International Sustainability Standards Board (ISSB), which merges elements of SASB and IFRS. ESG measurement is maturing, and the direction is clear: greater consistency, accuracy, and transparency. The relevance of ESG goes beyond compliance or branding. It represents a deeper shift in how companies are expected to operate. ESG is becoming a strategic imperative for long-term survival and is anchored in the concept of “social license to operate”. As externalities intensify,

from greenhouse gas emissions to supply chain disruptions, public expectations and stakeholder scrutiny are growing. Companies that fail to meet these expectations risk losing credibility, customers, and even access to capital. At the same time, leading firms show that ESG can be more than just risk mitigation. It can drive innovation, build brand trust, and unlock new growth. Firms like Patagonia and Natura &Co demonstrate that it's possible to generate both social impact and long-term profitability by including ESG into their purpose and operations. The ESG maturity curve offers a roadmap for companies to evolve. From compliance-driven efforts to integrated, purpose-led strategies that shape business decisions and stakeholder engagement.

### **Navigating ESG**

The article **“How to Make ESG Real”** from McKinsey emphasizes that while ESG may evolve in terminology or structure, its core purpose, helping companies manage rising externalities and maintain their social license to operate, is here to stay. The key is not whether to pursue ESG, but how to do that strategically, systematically, and authentically. Forward-looking companies, which are those that integrate ESG into core strategy, understand that they can't excel in every aspect of ESG. Instead, they define focused priorities, identify their strengths and vulnerabilities, and build tailored portfolios of initiatives that match their business models and purpose. ESG is framed not as an outcome but as an ongoing, evidence-based process.



**Mapping:** Understanding stakeholders, superpowers, and vulnerabilities.

In-depth ESG mapping reveals influential stakeholders and their values, with explicit corporate purpose facilitating ESG integration. Progressive companies involve shareholders but also hear out a wide array of stakeholders. Employees are essential because their engagement increases loyalty and advocacy. Consumers increasingly expect firms to be socially and environmentally responsible, with Gen Z particularly caring about ethical behavior, transparency, and authenticity. Purpose-led brands are reportedly delivering more than double the brand-value growth of profit-driven ones. The second aspect of ESG mapping is mapping a company's signature "superpowers", specific strengths for impact, and "vulnerabilities", stakeholder expectations attached to its business model. It entails asking the question of what makes the company indispensable, where its practices require synchronization, and how to utilize "home field" advantages, such as Natura &Co's Amazon-based biodiversity emphasis. Proactive companies perform "ESG teardowns", stringent examinations of existing efforts, in order to steer clear of imitations and concentrate on where they can set the pace. They map out ESG initiatives along their greatest strength, e.g., pharma on accessibility, renewables on emissions, or food companies on both environmental and social scores. Deep ESG analysis reveals risks, opportunities, and areas of underperformance. Firms need to evaluate how far ESG is integrated into governance, operations, budgeting, and stakeholder engagement, with strategy being translated into action with substance. Forward-thinking firms carefully choose ESG metrics and peer groups, typically informed by academics and thought leaders. Research indicates that companies with a stakeholder focus perform better operationally in the long run. Instead of adhering to traditional sector comparisons, these firms use wider peer benchmarks that mirror

how stakeholders—such as potential employees—assess companies across regions and sectors and adjust analysis accordingly.

**Defining: Making strategic ESG choices.**

*Firms have two central ESG choices:*

**High jumps** - exceeding evolving societal demands in addition to compliance with the law, for example, living wages, net-zero ambitions, and diversity requirements.

**Long jumps** - picking one or two ESG spaces to lead in, using distinctive strengths to set their ecosystem abuzz.

High and long jumps are distinct from recognizing superpowers and weaknesses; they're the doable actions that companies undertake. For instance, Maersk is the driving force behind zero-carbon shipping, while Walmart leverages its enormous supplier base for Project Gigaton, a mission to reduce a gigaton of emissions by 2030. Forward-looking businesses openly weigh ESG trade-offs, weighing costs and benefits—such as the dangers of inaction. For instance, increased employee compensation appears to be expensive in the short run but tends to increase productivity, trust, and competitive edge. But there are only so many resources, so an investment in one can affect another. These firms thrive by weighing carefully, making intentional decisions, moving definitively, and calculating the changing costs of not participating in key ESG areas. Making ESG tangible involves quantifying what really counts—targeting essential metrics consistent with long-term aspirations and business strategy. Successful ESG performance management establishes clear milestones, monitors the right KPIs, and connects targets to essential operations. Monitoring on a regular basis, backed by robust data analytics, allows companies to respond rapidly to changing circumstances and remain poised in stable or fast-changing situations.

**Embedding: Aligning ESG with operations.**

While ideally corporate purpose should come prior to ESG strategy, in the real world, firms first address immediate operating challenges, as was the case with Best Buy's revival under Hubert Joly. This doesn't necessitate neglecting ESG. Effective firms embed ESG within the core strategy through 5Ps: portfolio and products, people and culture, processes and systems, performance metrics, and external positions and engagement. By connecting these with stakeholders and their business model, they deliver strong, long-term ESG performance. Forward-looking businesses incorporate ESG into strategy and follow through, frequently with significant stakeholder backing. Patagonia is an example of this by making environmental commitments integral to its business operations. Incentives such as connecting executive compensation to ESG targets drive responsibility. In addition to financial rewards, companies employ behavioral "nudges" and open progress reporting to get employees on board and make ESG a business imperative every day, rewarding teams and individuals for driving these objectives. Leading companies improve their ESG metrics over time and choose to follow two or three ratings agencies that are strategic to them, rather than falling into the trap of pursuing scores at the expense of objectives. ESG standards are advancing worldwide—IFRS and ISSB now regulate international reporting, with IOSCO governing implementation. The EU is further developing its own standards of reporting. While frameworks are converging, pending challenges include creating impact-weighted accounting to reflect financial, social, and environmental performance in its entirety.

**Engaging: Communicating with stakeholders and investors.**

Forward-looking businesses treat communication as a means for learning, change, and engaging with various constituents, particularly employees. They learn from trade associations and

partnerships as well. Open engagement enables businesses to withstand criticism without belligerence. For example, Dick's Sporting Goods was criticized after stopping sales of assault-style weapons but remained open to conversation, stood by its purpose, and ultimately exceeded previous earnings and stock value. Investors are more and more demanding transparency and accountability regarding ESG, anticipating clear connections between ESG actions and corporate strategy. Progressive companies monitor quantifiable ESG results, embed sustainability within reports, and strengthen investor relations through specialized teams and data monitoring. Others also raise capital from green or sustainability-linked bonds (SLBs) tied to particular ESG targets, such as Burberry and Novartis raised in 2020. No matter the funding means, more disclosure is what creates stakeholder trust and sets companies up for success despite increasing global ESG regulations. Forward-looking businesses understand timely ESG disclosure is as important as the quality and substance of it. Regular reporting demonstrates commitment and gets better with time, integrating ESG into day-to-day business. Similar to financial information, easily accessible ESG data allows companies to act quickly, even in times of crisis, making ESG an ingrained business value.

Ultimately, the article argues that companies must move beyond superficial ESG efforts. Making ESG real means adopting a rigorous and evidence-based approach. Forward-looking companies define ESG by their business model and engage continuously with stakeholders. They treat ESG not as a temporary trend but as an enduring strategic necessity.

### **The Globalization of ESG Trends**

Since 2022, a number of economic difficulties have forced businesses to put survival first, including a worldwide recession, unstable supply chains brought on by the conflict between Russia and Ukraine, and an increase in corporate bankruptcies. Large-scale protests have resulted

from growing opposition to carbon neutrality initiatives in Europe. Anti-ESG attitude has increased in the United States, as seen by the 271 anti-ESG measures that were submitted and the 35 that were passed into law between 2022 and 2023. In light of this, investments in ESG have also decreased. ESG investments decreased by \$5 trillion to \$30.3 trillion in 2022 from \$13.3 trillion in 2012 to \$35.3 trillion in 2020, according to the Global Sustainable Investment Alliance (GSIA).

Examining the main ESG-related conversations that have emerged from recent significant international conferences and exhibitions is essential given the changing landscape of ESG management. Businesses must prioritize and concentrate on a wide range of ESG issues; therefore, it's critical to look at which issues are becoming more popular and how important stakeholders are reacting. The main ESG themes from three significant international events that took place between late 2024 and early February 2025 are highlighted in this article. These three gatherings, which provide critical viewpoints on ESG advancements, are among the most reputable international forums in the last three months.

The 29th United Nations Climate Change Conference (COP29), which will take place in Baku, Azerbaijan, from November 11 to 24, 2024, is the inaugural event. Based on the 1992 United Nations Framework Convention on Climate Change (UNFCCC), the Conference of the Parties (COP) is an intergovernmental gathering that brings together almost all nations to address climate action. The 2015 Paris Agreement, which aimed to keep global warming to 1.5°C, was a notable outcome of COP21 and acted as a spur for the development of ESG-focused corporate strategies. COP29 is an important event to examine since the agreements reached there will have a big impact on business climate action programs as well as national legislation.

CES 2025, which takes place in Las Vegas, USA, from January 7–10, is the second important event. One of the most important conferences in the world's IT sector, CES addresses urgent global issues while influencing future technical trends. In addition to showcasing state-of-the-art inventions, CES is essential for articulating a vision for sustainable technology and how it may address environmental and societal problems.

The World Economic Forum (WEF) 2025, which takes place in Davos, Switzerland, from January 20 to 24, is the last event. WEF is a key forum for high-level debates on social, political, and economic concerns and is crucial in establishing the global ESG agenda.

Even though each event had a unique focus, four important ESG issues showed up as a recurring theme. The focus on energy transition is the first. Discussions centered on the necessity for climate action and the energy equity issues brought on by the conflict between Russia and Ukraine, emphasizing the value of expanding renewable energy sources and maintaining a diverse energy mix. The second is using technological innovation to solve problems. It was widely acknowledged that developing new technologies is crucial to creating a future that is more ecologically friendly. Among the subjects covered were closing the digital divide and enhancing energy efficiency with digital technologies.

The third is increasing the involvement of the private sector in climate initiatives. Both public and private investment must be aggressively increased in order to promote climate change mitigation and adaptation. The contributions of rich nations have historically received more attention, but multilateral development banks and private businesses are now being highlighted more and more. Addressing the difficulties brought on by the development of AI is the last concern. Adoption of AI increases convenience, but it also raises ethical questions, increases

power demand, and widens digital inequality. The necessity to identify these problems and create suitable solutions is becoming more and more pressing.

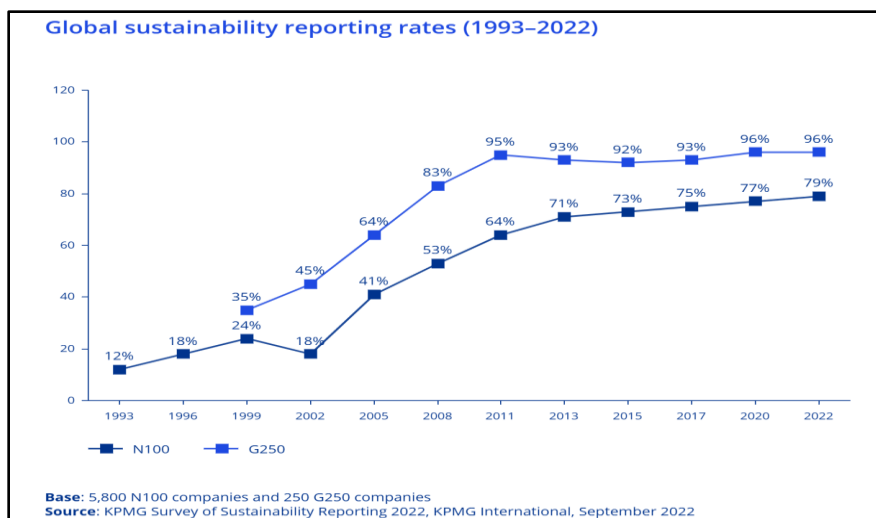
***Major issues in ESG in line with global trends:***

**This table is taken from the Samsung SDS Insights Report (available at [Insights | Samsung SDS](#))**

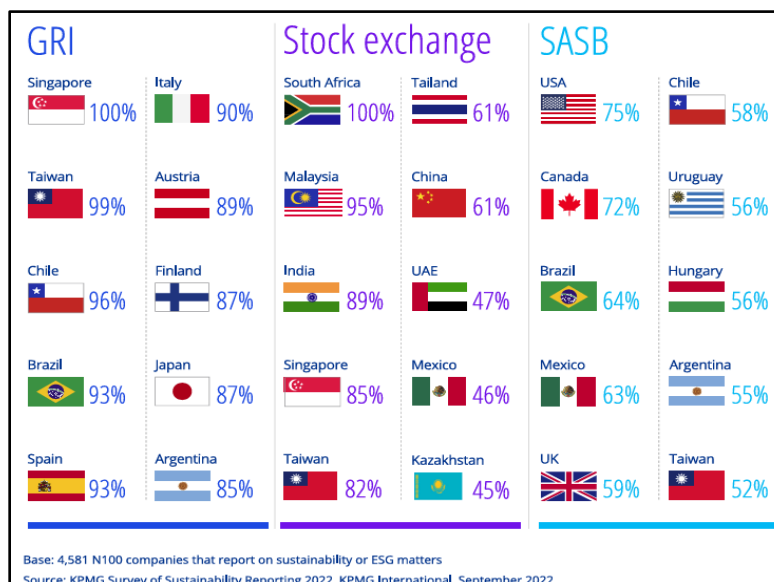
|                   | COP29                                                                                                                   | CES                                                                                 | WEF                                                                                                                          |
|-------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Energy Transition | agreed-upon objectives for the energy transition, such as doubling energy efficiency and tripling renewable power.      | held the Energy Transition show, released energy, and displayed connected products. | solutions for optimization The trilemma's highlighted remedies are affordability, sustainability, and energy security.       |
| Tech Innovation   | created the Digitalization Day events with a focus on utilizing technology to reduce greenhouse gas emissions and adapt | highlighted energy-saving technology to combat AI's growing power consumption       | suggested technologically based remedies for global issues like the digital divide and technology-assisted talent retraining |

|                    |                                                                                                                                   |                                                                                                        |                                                                                                                                                                        |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | to climate change.                                                                                                                |                                                                                                        | (EdTech)                                                                                                                                                               |
| Private Engagement | Establish worldwide goals, such as funding at least \$1.3 trillion (about KRW 1,911 trillion) by 2035.                            | showcased a range of green technologies from private businesses to combat the climate catastrophe.     | emphasized the value of private investment in the energy transition and encouraged private sector involvement.                                                         |
| AI                 | Identified problems and developed solutions for the surge in electricity demand brought on by the use of AI and cloud technology. | Technologies for resolving ethical and legal concerns brought on by AI developments were demonstrated. | AI-based energy optimization must strike a balance between the risk and opportunity of rising electricity demand and hastening the switch to renewable energy sources. |

According to KPMG's survey, the N100 companies' reporting rates have been rising consistently with every worldwide survey. Sixty-four percent of N100 businesses reported ten years ago. Seventy-nine percent of these businesses report in 2022. Nowadays, almost all of the top 250 firms in the world (G250) publish sustainability reports. The 96 percent reporting percentage among the G250 in 2022 is unchanged from 2020.



Since 2017, the percentage of N100 firms that include sustainability information in their annual financial reports has stable, with 60% of companies reporting in 2022. However, we observe a drop of 8 percentage points to 68 percent among the G250 after 2020. The Asia Pacific area is notably well-represented among the N100, accounting for six of the top ten nations, territories, and jurisdictions. Thirty percent of N100 firms in Asia Pacific and fifty-five percent of N100 corporations in the Middle East have adopted integrated reporting, which is a report that integrates both financial and non-financial data in a single annual report.



The above figure shows the top 10 countries, territories, and jurisdictions by percentage of N100 companies reporting against GRI standards, stock exchange guidelines, and SASB standards (2002).

The Sustainability Accounting Standards Board (SASB), the Global Reporting Initiative (GRI), and national stock market regulations are the three main reporting standards that were examined in the KPMG survey. Adopted by 68 percent of the N100 and 78 percent of the G250, the GRI is still the most widely used standard globally, with the Americas showing the highest adoption rates. Almost 25% of the N100 and G250 follow the rules or norms of their home stock exchange. The Middle East, Africa, and Asia Pacific areas have especially high adoption rates; China has the highest reporting rate at 64%. Corporations in the US and Canada are the main reasons behind the more than half of American corporations that report against SASB criteria. Outside of the Americas, SASB standards are becoming more and more popular; among Europe's N100, 35% of them have been adopted.

Initiatives such as the SFIA ASEAN Safe Platform, the Monetary Authority of Singapore's (MAS) Gprnt, and the free ESN Asia-Pacific Green Deal Assessment on ESGpedia have set the standard for enabling SMEs in Asia to begin their sustainability journey. In this trend, Europe is not far behind. A clear acknowledgement of the significance of SMEs in the larger fabric of a sustainable future, the European Commission's commitment to a SME-friendly environment was reiterated in March 2020 with the release of the "An SME Strategy for a Sustainable and Digital Europe."

Global sustainability reporting has advanced significantly in 2024, demonstrating a concerted effort to increase accountability and transparency. With countries from more than half

of the world's economy adopting its guidelines, the International Sustainability guidelines Board (ISSB) has played a crucial role in establishing a global standard for sustainability reporting. The interoperability initiatives between the Taskforce on Nature-related Financial Disclosures (TNFD) and the Global Reporting Initiative (GRI), which seek to simplify reporting standards and lessen market fragmentation, complement this approach. Global stakeholders gathered in Milan on October 18th for the 2024 IFRS Foundation Integrated Thinking and Reporting Conference to talk about the importance of integrated reporting and how it supports the ISSB's IFRS® Sustainability Disclosure Standards (IFRS S1 and IFRS S2). A new guidance was published by the IFRS Foundation to assist businesses in identifying and disclosing important sustainability-related risks and opportunities that may have an impact on their cash flows, cost of capital, or capacity to obtain financing. Understanding the IFRS S1 concept of sustainability risks and opportunities—including how they stem from a company's dependencies and impacts—is emphasized throughout the guide. Additionally, it emphasizes the materiality process, demonstrating how businesses can efficiently apply ISSB criteria by utilizing current financial statement preparation procedures.

In order to impose due diligence on human rights and environmental implications, the European Union (EU) has also been active, creating the Corporate Sustainability Due Diligence Directive (CSDDD). The quality and comparability of sustainability disclosures are also intended to be improved by the EU's dual reporting obligations under the IFRS Sustainability Disclosure Standards and the European Sustainability Reporting Standards (ESRS). With effect from January 5, 2023, EU member states must incorporate the Corporate Reporting Directive (CSRD) into their national laws within 18 months. By requiring more comprehensive and in-depth disclosures under the EU Sustainability Reporting Standards (ESRS), this directive builds upon

the Non-Financial Reporting Directive (NFRD). As of the year ending December 31, 2024, some major European companies are already required to make climate reports. In order to improve the comparability and dependability of these ratings and boost investor confidence in sustainable financial products, the European Council established new rules for ESG rating companies.

The Securities and Exchange Commission (SEC) in the US finalized its climate disclosure regulations, which require thorough disclosures about climate change in financial reporting. When taken as a whole, these programs represent a major step toward a more unified and efficient global sustainability reporting environment, inspiring companies everywhere to implement strong and open sustainability policies.

The reporting agenda in Australia is dominated by climate issues, with governance and social issues coming in second and third. Corporate responsibility continues to be the most often disclosed topic. Significant progress in sustainability reporting was made in Australia in 2024, signaling a turning point in the direction of increased corporate responsibility and openness. In accordance with the International Sustainability Standards Board's (ISSB) recommendations, the Australian Accounting Standards Board (AASB) has authorized new sustainability reporting standards. With a phased implementation for smaller companies, these guidelines, which go into effect on January 1, 2025, require climate-related disclosures from major entities. To assist companies in getting ready for these new regulations, the Australian Securities and Investments Commission (ASIC) has released recommendations. Strong governance and record-keeping procedures are emphasized in the guidelines. With a staggered implementation timetable that corresponds with the AASB's S2, the federal and state governments have also introduced comparable standards for the public sector, guaranteeing uniformity between the public and private sectors. A staged method to auditing these disclosures is also outlined in the suggested

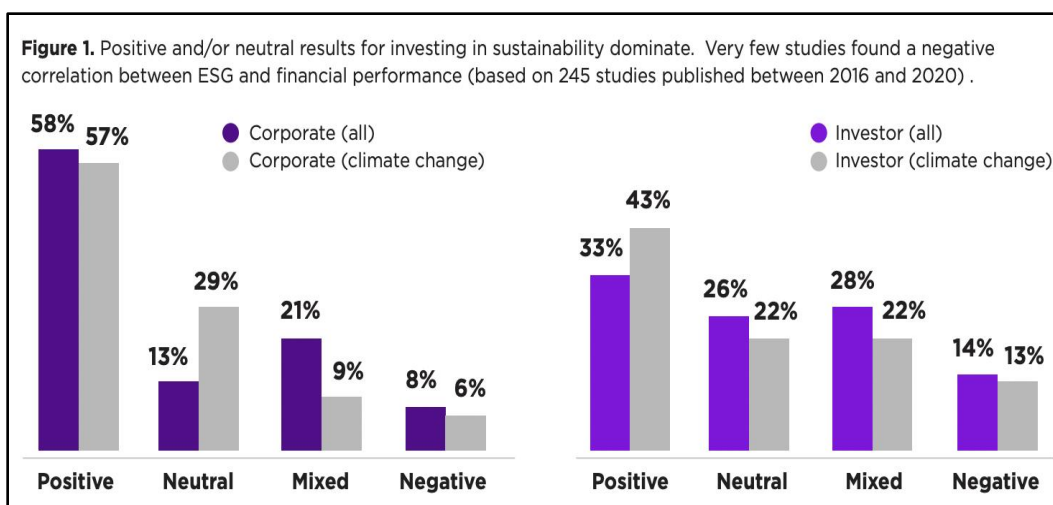
schedule for assurance over sustainability reports, which begins with low assurance and advances over a number of years to reasonable assurance. The goal of this staged approach is to allow auditors and businesses time to build the skills they need. In order to stay in compliance and take advantage of the chance to improve their sustainability processes, Australian firms must immediately adjust to the new requirements when mandatory climate reporting takes effect in January 2025.

### **Empirical Evidence on ESG Disclosures and Financial Performance**

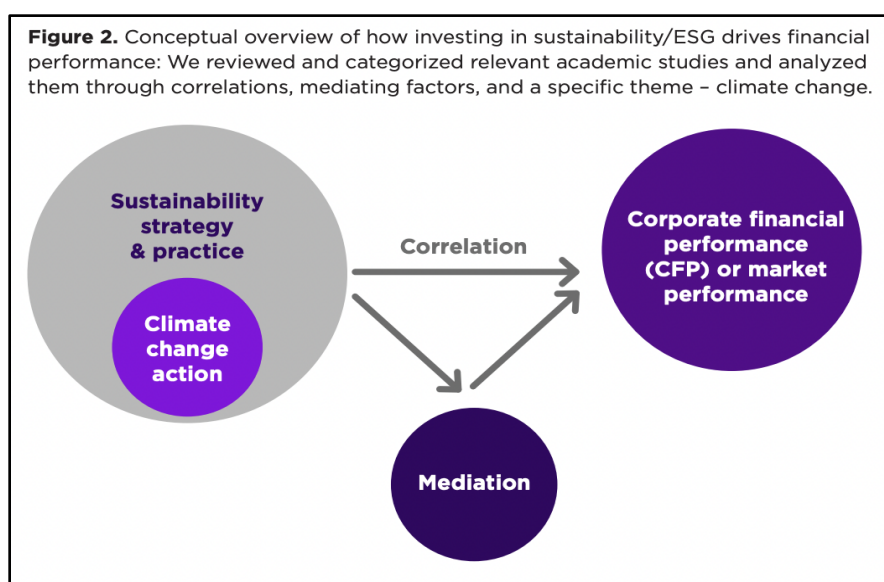
The ESG Institute report underscores a fact that over 80% of the investors agree that sustainability and financial performance are not trade-offs.

Empirical research indicates a generally positive relationship between ESG disclosures and financial outcomes. Broadly speaking, ESG disclosures influence investor behavior by mitigating risk and enabling more informed capital allocation. However, the impact on financial performance is not driven by disclosure alone but instead it depends on the strategy and depth behind what is reported.

Bentleys emphasizes that effective ESG reporting plays a critical role in business strategy and risk management. Companies that follow structured ESG disclosure frameworks such as GRI, SASB and TCFD not only strengthen their regulatory compliance but also enhance investor confidence. ESG disclosures around emissions, labor policies, board governance and climate risk enable stakeholders to assess long-term sustainability and financial resilience. According to Bentleys, such disclosures contribute to stronger reputations, better stakeholder trust and reduced exposure to regulatory or reputational risks.



*The NYU Stern-Rockefeller analysis found that 58% of corporate-focused studies demonstrated a positive link between ESG practices and financial metrics like ROA, ROE or stock price.*



**However, the NYU Stern-Rockefeller meta-study presents a critical distinction:** disclosure by itself does not necessarily lead to better financial performance. Among the 1000+ studies reviewed between 2015 and 2020, only 26% of those focusing on disclosure alone reported a positive financial correlation, compared to 53% for studies examining ESG performance outcomes such as carbon reductions or governance reforms. ESG metrics that reflect genuine

operational improvements rather than superficial compliance are more likely to signal long-term value creation to investors.

For investor-focused studies, 59% showed ESG portfolios performed as well or better than conventional ones, particularly when ESG integration was strategic. Moreover, climate-focused strategies like decarbonization or low-carbon investing tended to outperform peers. They offered not just returns but also downside protection during crises like the COVID-19 pandemic and the 2008 financial crash. These trends underscore how ESG disclosures can improve risk-adjusted returns and investor confidence when paired with meaningful initiatives. ESG-aligned companies often show more stable earnings and greater access to capital.

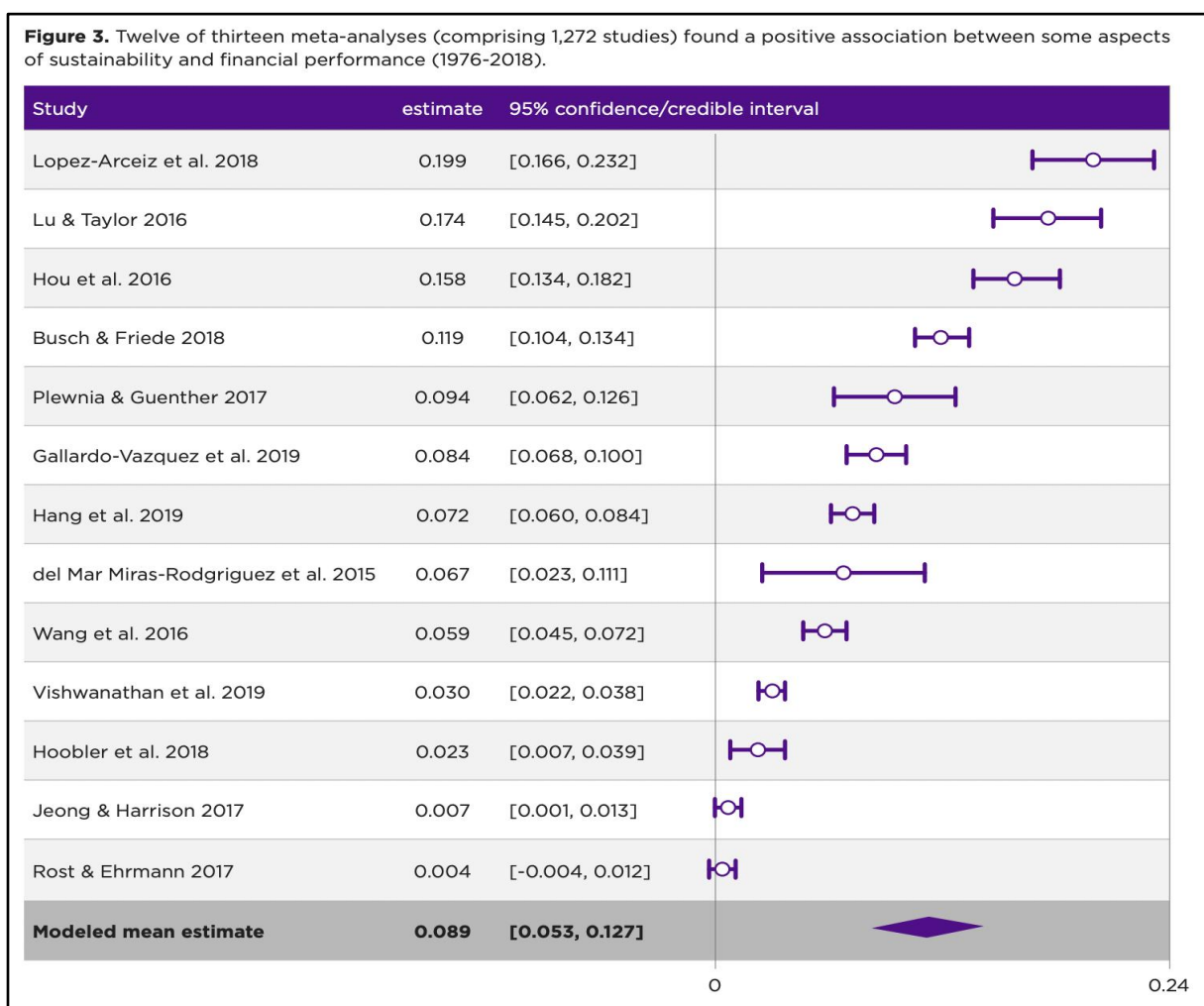
The financial benefits of ESG disclosures are often amplified by mediating factors such as risk mitigation, innovation, and operational efficiency. According to the NYU Stern-Rockefeller report, studies that incorporated these factors were significantly more likely to find positive financial outcomes. For example, portfolios with lower ESG risk exposure were more resilient and better positioned for long-term outperformance.

Moreover, ESG integration was found to outperform simpler strategies like negative screening. ESG Improvers (firms showing consistent progress on material ESG issues) delivered annualized outperformance of 3.8% compared to ESG Decliners. This highlights that dynamic ESG progress often signals stronger management and adaptability, which are attributes valued by long-term investors.

Both Bentley and the NYU Stern-Rockefeller study highlight the importance of focusing on material ESG issues. Generic or irrelevant disclosures reduce impact, whereas ESG metrics aligned with sector-specific risks and opportunities are more likely to have an effect on investor

decision-making. For instance, disclosures tied to climate risk, board diversity, or supply chain ethics tend to carry greater weight in sectors where those risks are financially significant.

Investors increasingly differentiate between ESG “leaders” and “improvers”, with the second showing untapped potential. The research suggests that it is not just where a firm currently stands on ESG, but the trajectory of improvement that matters most.



A long-term ESG focus raises the probability of favorable or neutral financial results by 76%, according to studies. According to Hang et al. (2019), environmental investments provide long-term benefits but no immediate financial impact. According to Kotsantonis et al. (2019),

when a CEO communicates long-term intentions, the market responds favorably. According to Dorfleitner et al. (2018), companies with high ESG ratings see mid- and long-term returns that are up to 3.8% higher per standard deviation.

Due to different methodologies and small sample numbers, research on ESG investment yields inconsistent results. The majority of research focuses on stocks, where 33% of studies demonstrate positive alpha; fixed income and passive methods provide less convincing evidence. Though pooled ESG techniques have less compelling results, probably because of heterogeneous approaches, ESG integration generally performs better than negative screening and divesting. Few studies differentiate between ESG leaders and improvers or highlight significant ESG concerns. The alpha potential of ESG Improvers is highlighted by Rockefeller Asset Management and Khan et al. (2016). A fictitious portfolio of the best ESG improvers beat the ESG decliners by 3.8% a year between 2010 and 2020. When paired with elements like quality, low volatility, value, and momentum, optimized ESG improver portfolios also increased returns, indicating that ESG enhancement offers quantifiable investing value.

Asymmetric benefits are typically provided by ESG investing, particularly in terms of lowering risk and improving financial performance during recessions. Research indicates that during the 2008 crisis, German green funds and FTSE4Good indices did better than their counterparts. Better risk-adjusted returns during downturns were repeatedly demonstrated by highly rated ESG mutual funds (Chatterjee, Das). 24 out of 26 ESG index funds beat traditional ones during the COVID-19 Q1 2020 meltdown, and by Q3, they were still outperforming by 45% (Morningstar). Although Demers et al. (2020) noted an exception, the majority of investigations support this link.

According to the ROSI framework, corporate sustainability can improve financial performance through mediating elements like innovation, operational effectiveness, risk management, and stakeholder interactions. According to Vishwanathan et al. (2019), the main motivators are innovation, risk mitigation, stakeholder reciprocity, and reputation. Despite having small sample sizes, all 17 research pertaining to innovation demonstrated a favorable financial benefit. Of the 22 investigations, 59% indicated a favorable correlation with operational efficiency. Risk was crucial; compared to 48% of investor studies that included risk, only 27% of those that did not had favorable findings. A positive association was confirmed in roughly 52% of all risk-focused research. Stronger financial results were also associated with improved climate risk management (51% positive), according to studies. However, because of small samples and overlapping effects, no single component produced statistically significant outcomes across all models, suggesting further research in this field.

Although leading finance journals have not yet published on the subject, research on decarbonization tactics demonstrates a high correlation with enhanced financial success. The majority of the 59 research that were analyzed produced favorable findings, particularly for tactics that aggressively reduced carbon emissions. While In, Park, and Monk (2019) demonstrated that carbon-efficient enterprises produced annual anomalous returns of 3.5% to 5.4% without the need for government incentives, Cheema-Fox et al. (2019) proved that such techniques produced greater risk-adjusted returns. Although there are currently few studies on climate solution investments, they show promise because it is anticipated that regulations, consumer behavior, and technology would affect markets as a result of climate change. This is supported by the fact that, from 2015 to 2020, the FTSE Opportunities All Share Index beat the FTSE Global All Cap Index by 4.9% yearly.

Compared to 53% for performance-based ESG metrics, only 26% of studies that focused on disclosure shown a positive correlation with financial performance. This implies that disclosure by itself has little effect in the absence of tangible ESG action. For example, because UN PRI signatories prioritized disclosure over performance, neither ESG nor financial outcomes increased (Kim & Yoon, 2020). Fatemi et al. (2018) also discovered that while simple disclosure may have a detrimental impact on valuation, strong ESG performance increases firm value.

An excellent illustration of how incorporating ESG principles can result in better financial outcomes is Unilever. Significant financial gains have resulted from the company's Sustainable Living Plan, which attempts to lessen its environmental effect while enhancing its positive social impact. Brands that supported Unilever's sustainability strategy grew 69% faster than the company as a whole and accounted for 75% of its total revenue, according to the company's 2020 annual report. This illustrates how giving ESG top priority may boost sales, improve brand loyalty, and eventually improve financial performance.

The outdoor apparel brand Patagonia offers yet another interesting case study. The business has centered its brand on social responsibility and environmental sustainability, placing a strong emphasis on openness and moral behavior. Patagonia has built a devoted following of customers and a reputation as a pioneer in corporate responsibility by embracing a strong commitment to ESG principles. A 2019 Harvard Business Review research claims that in addition to boosting sales, Patagonia's commitment to sustainability has protected the business from the harm that unsustainable practices could do to its brand. Over the last ten years, the company's revenues have increased by 10–20% annually, which reflects the financial gains from its ESG initiatives.

## Challenges and Criticisms

Green washing can skew stakeholders' opinions of corporate sustainability initiatives and poses a serious threat to the legitimacy of ESG reporting. To gain the trust of investors and customers, businesses must implement honest and open reporting procedures. The corporate community may promote a more genuine ESG environment where sincere sustainability efforts are acknowledged and rewarded by combating greenwashing, eventually promoting constructive change in business performance and behavior.

Concerns about comparability and data inconsistency are serious obstacles in the field of Environmental, Social, and Governance (ESG) reporting. The absence of uniform metrics and reporting structures, which might differ significantly between sectors and geographical areas, is the root cause of these difficulties. Because of this, stakeholders frequently struggle to accurately assess and contrast ESG performance.

Corporations, especially smaller ones, frequently face substantial cost difficulties when implementing Environmental, Social, and Governance (ESG) programs. Even while the long-term advantages of ESG compliance—like improved reputation and lower risks—are well known, the upfront expenses might be prohibitive. Even while pursuing ESG measures can pay off in the long run, the related costs are a major obstacle, especially for smaller organizations. Promoting wider adoption of sustainable practices will need addressing these financial obstacles with focused assistance and resources.

The continuously changing landscape of ESG laws poses serious difficulties for firms trying to meet a variety of standards. Businesses must comply with a growing number of complicated rules that differ by location and industry as governments and regulatory agencies throughout the world place a greater emphasis on sustainable business practices. The absence of

uniformity in ESG reporting systems is a significant obstacle. Because different countries have different rules, it is challenging for businesses to standardize their reporting procedures. For example, businesses that operate globally can find it difficult to adhere to the U.S. Securities and Exchange Commission's (SEC) changing ESG disclosure regulations as well as the European Union's Sustainable Finance Disclosure Regulation (SFDR) (Eccles & Klimenko, 2019). This discrepancy raises the administrative load and related expenses in addition to making compliance more difficult. Furthermore, because ESG standards are ever-changing, businesses need to be alert and flexible. As companies strive to adapt their operations to changing regulatory environments, the changing standards for accountability and transparency may cause ambiguity. As a result, businesses can find it difficult to guarantee that their ESG plans are both efficient and legal, requiring constant expenditure of funds and knowledge.

The most common criticism of ESG has probably been that it interferes with what detractors believe to be the core of what companies should do, which is to "make as much money as possible while conforming to the basic rules of the society," as Milton Friedman put it more than fifty years ago. According to this viewpoint, ESG might be portrayed as a sideshow, a PR stunt, or even a way to capitalize on the more noble motivations of clients, investors, or staff. ESG is "good for the brand," but it isn't the cornerstone of business strategy. It is sporadic and additive. For instance, MSCI, a provider of ESG ratings and scores, discovered that almost 60% of 2021's "say on climate" votes were one-time affairs, with less than one in four of these votes having yearly follow-ups planned. ESG initiatives have been criticized by some as "greenwashing," "purpose washing," or "woke washing." For instance, according to a poll conducted by Edelman, almost three out of four institutional investors do not believe that

businesses will fulfill their declared commitments to sustainability, ESG, or diversity, equality, and inclusion (DEI).

Another criticism of ESG is that, in addition to fulfilling the technical specifications of the E, S, and G components, it is just too difficult to find the balance needed to use ESG in a way that appeals to a variety of stakeholders. Maximizing value for the company and its shareholders is the obvious goal while solving for a financial return. However, what if the scope is expanded and the workable solutions are far more intricate? It can be difficult or even impossible to find a solution that works for all of the stakeholders. Who should management give the extra ESG dollar to? In the form of reduced costs for the consumer? To the workers, through better pay or more benefits? To vendors? Maybe by an internal carbon tax, in the direction of environmental issues? The best option isn't always obvious. Furthermore, it is uncertain whether a firm would have a clear mandate from its shareholders to make such a decision even if it were possible.

Another argument is that it is impossible to measure ESG, especially as it is represented in ESG scores. Some detractors contend that aggregate ESG scores are meaningless, even though individual E, S, and G characteristics can be evaluated if the necessary, auditable data are collected. Disparities in methodology and weighting among ESG rating and score providers exacerbate the shortcoming. ESG scores from six of the most well-known ESG ratings and scoring providers, for instance, correlate on average by just 54% and range from 38% to 71%, whereas the credit scores of S&P and Moody's correlated at 99%. Additionally, groups like the Sustainability Accounting Standards Board (SASB) and the Global Reporting Initiative (GRI) might assess the same phenomenon in different ways. For instance, SASB counts by training hours, but GRI examines staff training in part by amounts invested in training. Divergent scores from various rating and scoring providers, each of which uses their own weightings and analyses,

are therefore to be expected. Furthermore, big investors frequently employ their own proprietary approaches that are refined over time and incorporate a range of inputs, including ESG rankings.

One more criticism of ESG is that, when they do occur, favorable correlations with outperformance could be attributed to other factors and are not causal. If ESG ratings from multiple ratings and scoring providers, measuring different industries, employing different methodologies, weighting metrics differently, and looking at a variety of companies operating in different regions all produced nearly identical scores that nearly exactly matched company performance, then it would certainly defy logic. Performance correlations are prone to change and can be explained by a variety of factors, such as industry headwinds or tailwinds. Any causal relationship between ESG performance and financial performance has been questioned by a number of researchers. Even though a recent meta-study found that most ESG-focused investment funds do beat the overall market, 20% of them do not, and even the companies and funds that have outperformed may have another reason for doing so. (Technology and asset-light companies, for instance, are frequently among the industry leaders in ESG ratings; their relatively minimal carbon footprint makes them deserving of better ESG scores.) In a recent study, the director declared emphatically that "there is no ESG alpha."

In addition to these criticisms, some have questioned whether ESG ratings are still relevant in light of current events and volatile markets. It is true that the acknowledged, urgent need to improve energy security following the invasion of Ukraine may, in the short term, result in increased extraction and use of fossil fuels, and that the war and its aftermath may jeopardize the international cooperation necessary for a more orderly net-zero transition. Additionally, tolerance with what could be referred to as "performative ESG" rather than "true ESG" is

probably going to run thin. A prudent, well-thought-out approach that supports a company's mission and business plan is consistent with true ESG (exhibit).

### **Future of ESG**

There is more to sustainable finance than a fad. ESG measurement is becoming a crucial component of contemporary finance due to a number of factors, including more precisely defined regulatory frameworks, rising investor demand, ethical developments, and technical breakthroughs that improve its accuracy. This change reflects a more comprehensive knowledge of how social, economic, and environmental well-being are intertwined. This movement is expected to pick up speed as demand and awareness for sustainable investments continue to grow, bringing about major changes in the global economy and paving the way for a more sustainable future.

ESG is still a priority for Europe and Asia, and new laws are being implemented in these regions. Meanwhile, political opposition is a problem in the US and certain emerging markets. Anticipate a growing disparity between these areas, which is detrimental to the boards of transatlantic corporations. All of a sudden, they must make sure that their company satisfies two very different political regimes. Many businesses will see the need for fresh perspectives on executive teams and boards, especially in Europe and Asia. They can only expect to adhere to the new reporting requirements in this way. As a result, ESG-related training will become increasingly important.

ESG reporting is about to undergo a transformation thanks to blockchain and AI. Consider automated sustainability audits and real-time supply chain monitoring. Digital is the way of the future since it can simplify large operations and allow businesses to report with confidence and detail. How can a business invest in ESG without upsetting its anti-ESG

government? The solution is hushing, which is simply keeping silent about a certain topic, regardless of how passionate you are about it, out of concern that going public may draw undue criticism. The most divisive views on ESG will lead to increased investor action, whether businesses and governments like it or not. Aggressive activism can occasionally jeopardize a board's entire agenda, so they must be ready for it.

### **Conclusion**

In summary, ESG disclosures are now a fundamental component in driving investor behavior in international financial markets, but with a subtle effect. While investors increasingly expect transparency and normalized reporting, disclosure itself is not enough to assure favorable investor reaction. There is evidence that investors only reward those companies that disclose ESG facts but also exhibit solid ESG performance in support of long-term strategic objectives. Simple compliance-driven disclosures do not typically cause confidence or financial returns and can even have a neutral or negative impact on valuation in some instances.

Yet when ESG disclosures are accompanied by effective action such as risk reduction, innovation, operational effectiveness, and stakeholder dialogue, they build trust, lower perceived investment risk, and help create long-term value. Also, during periods of market doubt or crisis, companies with solid ESG reputations and open communication are more likely to show resilience, gathering investor attention. As international regulations move toward tougher ESG reporting standards, how well companies incorporate ESG performance into genuine, material disclosures will probably emerge as a defining feature in investor choices. Finally, ESG disclosures have maximum effect when they are based on sincere commitment and are integrated into the business's operational and strategic framework.

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